Email Marketing SaaS

Frontend Review, Backend Alignment & Step-by-Step Implementation Roadmap

Tech Stack: NestJS + PostgreSQL (Prisma/RLS) & BullMQ | React 19 + Vite + Tailwind **Audit Date:** October 2026 | **Status:** Architecture Verified

> **Key Takeaway:** The backend is ~85-90% complete with 14 active modules, full quota gating, and email tracking pipelines. The frontend has completed 100% of its foundational shell (Authentication, Layouts, ⌘K command palette, and trial billing states), but business feature pages (Contacts, Templates, Campaigns, Automations, Forms, Sending, and Billing Upgrade) are currently pending UI views.

1. What is DONE in Frontend

Area	Status	Implementation Details & Key Files
Auth Flow & Security	**100% Done**	• 5 full pages: Login, Signup, Forgot Password, Reset Password, Verify Email. • Persisted Zustand store (`auth.store.ts`) + automatic 14-day trial startup. • Axios client (`api.ts`) with **single-flight 401 refresh token queue** and error normalization. • Route protection guards (`RequireAuth`, `RequireGuest`, `RequirePlatformAdmin`).
Layout & Shell	**100% Done**	• Acelle-style sidebar navigation (Campaigns, Audience, Sending) in `sidebar.tsx`. • Header with user profile menu, avatar initials, and fast logout. • Global Command Palette (⌘K / `Ctrl`+K) in `command-palette.tsx`. • Theme store & dark/light styling.
UI Component System	**100% Done**	• Radix UI + Tailwind primitives: Button, Dialog, Dropdown, Avatar, Badge, Checkbox, Select, Command, Input, Label, FormField. • Sonner toast notification system.
Reactive Trial Banner	**100% Done**	• `trial-banner.tsx` & `banner-state.ts` automatically monitor subscription state. • Triggers urgent amber warnings when ≤3 days remain, red banners on expired/past-due.
Customer Home Dashboard	**Skeleton Ready**	• `dashboard.page.tsx` with 3 Credit Meters (reading live subscription plan limits). • 4 7-Day Stat Cards (Total Sent, Open Rate, Click Rate, Bounce Rate) with layout verified.
Contacts Data Layer	**Hooks Ready**	• TanStack Query hooks in `use-contacts.ts` (useContacts, useContact, useCreateContact, useUpdateContact, useDeleteContact). • `status-badge.tsx` mapping Prisma status enum to visual badges.

2. What is PENDING in Frontend (Backend Gap Analysis)

Each item below corresponds to a fully developed backend controller waiting for frontend UI views:

Module & Route	Backend Endpoints Available	Pending Frontend Views & Logic

Audience: Contacts
`/app/contacts`
	`POST /api/contacts`
	`GET /api/contacts`
	`POST /api/contacts/import`
	`GET/PATCH/DELETE /api/contacts/:id`
	`POST/DELETE :id/tags`
		• Contacts DataTable with search, status filters, and pagination.
		• "Add Contact" modal dialog.
		• "Import CSV" drag-and-drop modal.
		• Contact details drawer & delete confirmation dialog.

Audience: Lists & Tags
`/app/lists`
`/app/tags`
	`GET/POST/PATCH/DELETE /api/lists`
	`POST/DELETE :id/contacts`
	`GET /api/lists/:id/contacts`
	`GET/POST/DELETE /api/tags`
	`GET/POST/DELETE /api/suppressions`
		• Lists management grid with contact counts.
		• List detail view showing assigned contacts.
		• Tags management view.
		• Suppression / Blacklist management screen.

Audience: Segments `/app/segments`	`GET/POST/PATCH/DELETE /api/segments` `GET /api/segments/:id/preview`	• Segments list screen. • Filter criteria builder (by tag, status, creation date). • Real-time audience size calculation preview.

Templates & Signatures
`/app/templates`
`/app/sending/signatures`
	`GET/POST/PATCH/DELETE /api/templates`
	`POST :id/duplicate`
	`GET/POST/PATCH/DELETE /api/signatures`
	`POST :id/verify, :id/default`
	`GET/POST /api/template-categories`
		• Template gallery grid with category filtering & preview.
		• HTML email template editor.
		• Sender Signatures screen ("From Name" & "From Email") with "Verify" and "Set as Default" buttons.

Campaigns & Delivery
`/app/campaigns`
	`GET/POST/PATCH/DELETE /api/campaigns`
	`POST :id/duplicate`
	`POST :id/schedule`
	`POST :id/pause, :id/resume, :id/cancel`
	`POST :id/resolve-recipients`
		• Campaigns list view with status indicators.
		• **4-Step Creation Wizard:**
		1. Details (Subject, Preheader, Sender)
		2. Audience (List or Segment)
		3. Template selection
		4. Schedule / Send Now button.
		• Actions: Pause, Resume, Cancel, Duplicate.

Campaign Analytics `/app/campaigns/:id`	`GET /api/campaigns/:id/analytics` `GET /api/campaigns/:id/analytics/timeline`	• Real-time delivery cards (Sent, Delivered, Opens, Clicks, Bounces). • Open and click rate metric indicators. • Hourly/daily activity timeline chart.
Sending Infrastructure `/app/sending/domains`	`GET/POST/DELETE /api/sending-domains` `POST :id/verify`	• Sending domains list with DNS status. • Add Domain modal displaying SPF, DKIM, and DMARC records. • "Verify DNS" button calling verification endpoint.

Module & Route	Backend Endpoints Available	Pending Frontend Views & Logic

Automations (Drip)
`/app/automations`
	`GET/POST/PATCH/DELETE /api/automations`
	`PUT :id/steps`
	`POST :id/activate, :id/pause, :id/enroll`
	`GET :id/runs`
		• Automation workflows list.
		• Step sequence editor (Trigger -> Wait -> Send Email -> Condition).
		• Workflow activation toggle & execution run log.

Forms & Lead Capture
`/app/forms`
	`GET/POST/PATCH/DELETE /api/forms`
	`GET :id/submissions`
	`POST /api/public/forms/:id/submissions`
		• Forms list & form builder.
		• HTML embed code modal.
		• Form submissions log viewer.
		• Standalone public form submission page.

Billing & Plans
`/app/billing`
	`GET /api/plans`
	`GET /api/billing/subscription`
	`POST /api/billing/subscription/checkout`
	`POST /api/billing/subscription/cancel`
		• Current subscription details & quota meters.
		• Pricing comparison cards (Starter, Pro, Business).
		• "Upgrade" button opening Razorpay checkout script.
		• Subscription cancellation modal.

Settings & Team
`/app/settings`
	`GET/PATCH /api/users/me`
	`PATCH /api/users/me/password`
	`GET/PATCH /api/tenants/current`
	`GET/POST/DELETE /api/users`
		• User profile and password settings.
		• Workspace / Company name settings.
		• Team member invitation & roles table.

3. Step-by-Step Implementation Roadmap

Follow this chronological 6-phase sequence to build out the screens in logical dependency order:

Phase 1: Audience & Contacts Management **START HERE**

Rationale: Without contacts and lists in your database, you cannot send campaigns or test automations.

- 1. Build **Contacts Table** (`/app/contacts`) using TanStack Table with search, status filters, and pagination.
- 2. Build **Add Contact Dialog** wired to `useCreateContact()`.
- 3. Build **Import CSV Dialog** with drag-and-drop CSV upload calling `POST /api/contacts/import`.
- 4. Build **Lists View** (`/app/lists`) and list subscriber view.
- 5. Register routes in `routes.tsx` for `/app/contacts` and `/app/lists`.

Phase 2: Templates & Sender Identities **Step 2**

Rationale: Every campaign requires an email body template and a verified sender signature.

- 1. Build **Sender Signatures** (`/app/sending/signatures`) to manage "From Name" and "From Email" with verify action.
- 2. Build **Template Gallery** (`/app/templates`) with thumbnail previews, categories, and duplicate action.
- 3. Build **Template Editor** (`/app/templates/:id/edit`) for HTML email content and subject lines.
- 4. Build **Sending Domains** (`/app/sending/domains`) with SPF/DKIM DNS verification instructions.

Phase 3: Campaigns Flow & Analytics **Step 3**

Rationale: This is the core revenue-generating product functionality of the platform.

- 1. Build **Campaigns List** (`/app/campaigns`) with status badges and action menus (Pause, Resume, Cancel).
- 2. Build **4-Step Creation Wizard**: Campaign Info -> Select List/Segment -> Select Template -> Schedule/Send.
- 3. Connect schedule API (`POST :id/schedule`) and recipient resolution (`POST :id/resolve-recipients`).
- 4. Build **Campaign Analytics Dashboard** (`/app/campaigns/:id`) with open/click rates and delivery timeline chart.

Phase 4: Billing & Upgrades **Step 4**

Rationale: The trial banner points to /app/billing; users must be able to upgrade before trial ends.

- 1. Build **Billing View** (`/app/billing`) showing current plan quota meters and trial countdown.
- 2. Render pricing tier cards from `GET /api/plans`.
- 3. Wire "Upgrade" button to `POST /api/billing/subscription/checkout` and load the Razorpay checkout modal.
- 4. Implement subscription cancellation confirmation dialog.

Phase 5: Lead Forms & Automations **Step 5**

Rationale: Drip automations and forms leverage your existing contacts, lists, and email templates.

- 1. Build **Forms Manager** (`/app/forms`) with embed code generator and submissions table.
- 2. Build public form landing page (`/public/forms/:formId`).
- 3. Build **Automations Workflows** (`/app/automations`) with trigger and step sequence configuration.
- 4. Add workflow execution run logs.

Phase 6: Workspace Settings, Team & Live Metrics **Step 6**

Rationale: Completes multi-tenant collaboration and connects live database stats to the home dashboard.

- 1. Build **Profile & Password Settings** (`/app/settings/profile`).
- 2. Build **Team Management** (`/app/settings/team`) for inviting team members and assigning roles.
- 3. Connect live counts into Home Dashboard Credit Meters (Lists, Campaigns, Subscribers).
- 4. Populate 7-day send volume and engagement rates.

4. Key Architectural Guidelines

- **Query Invalidation:** Follow the convention in `use-contacts.ts`: root invalidate on create/delete, detail + list invalidate on update.
- **Pagination Envelope:** Backend returns `{ data: T[], meta: { page, limit, total, pages } }`. Remember the key is `pages`, not `totalPages`.
- **Quota Gating Handling:** Backend returns `403 QUOTA_EXCEEDED` when tenant exceeds limits. Show a clean "Upgrade Plan" modal rather than a generic error toast.